

Roll. No.

O.M.R. Serial No.

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Question Booklet Number

600185

B.C.A. (SEM.-VI) (NON-NEP) EXAMINATION, 2024

COMPUTER APPLICATION

(Computer Network Security)

[BCA-601(N)]

Paper Code

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**Question Booklet
Series**

A

Time : 1 : 30 Hours

Max. Marks : 75

Instructions to the Examinee :

1. Do not open the booklet unless you are asked to do so.
2. The booklet contains 100 questions. Examinee is required to answer 75 questions in the OMR Answer-Sheet provided and not in the question booklet. All questions carry equal marks.
3. Examine the Booklet and the OMR Answer-Sheet very carefully before you proceed. Faulty question booklet due to missing or duplicate pages/questions or having any other discrepancy should be got immediately replaced.
4. Four alternative answers are mentioned for each question as - A, B, C & D in the booklet. The candidate has to choose the correct / answer and mark the same in the OMR Answer-Sheet as per the direction :

(Remaining instructions on last page)

परीक्षार्थियों के लिए निर्देश :

1. प्रश्न-पुस्तिका को तब तक न खोलें जब तक आपसे कहा न जाए।
2. प्रश्न-पुस्तिका में 100 प्रश्न हैं। परीक्षार्थी को 75 प्रश्नों को केवल दी गई OMR आन्सर-शीट पर ही हल करना है, प्रश्न-पुस्तिका पर नहीं। सभी प्रश्नों के अंक समान हैं।
3. प्रश्नों के उत्तर अंकित करने से पूर्व प्रश्न-पुस्तिका तथा OMR आन्सर-शीट को सावधानीपूर्वक देख लें। दोषपूर्ण प्रश्न-पुस्तिका जिसमें कुछ भाग छपने से छूट गए हों या प्रश्न एक से अधिक बार छप गए हों या उसमें किसी अन्य प्रकार की कमी हो, उसे तुरन्त बदल लें।
4. प्रश्न-पुस्तिका में प्रत्येक प्रश्न के चार सम्भावित उत्तर- A, B, C एवं D हैं। परीक्षार्थी को उन चारों विकल्पों में से सही उत्तर छंटना है। उत्तर को OMR उत्तर-पत्रक में सम्बन्धित प्रश्न संख्या में निम्न प्रकार भरना है :

(शेष निर्देश अन्तिम पृष्ठ पर)

SI

1. Which of the following refers to the violation of the principle if information/ computer is no more accessible?
 - (A) Availability
 - (B) Confidentiality
 - (C) Access control
 - (D) All of the above
2. In a computing organization _____ is a network security system that monitors and controls the incoming and outgoing network traffic based on predetermined security rules.
 - (A) spyware
 - (B) cookies
 - (C) spam
 - (D) firewall
3. A computer virus is a :
 - (A) software
 - (B) hardware
 - (C) bacteria
 - (D) freeware
4. Which of the following uses PGP application ?
 - (A) Email
 - (B) File encryption
 - (C) Email and file encryption
 - (D) None of these
5. _____ means a system of a secure key pair consisting of a private key for creating digital signature and public key for verifying digital signature.
 - (A) Asymmetric cryptography
 - (B) Cryptosystem
 - (C) Symmetric cryptosystem
 - (D) None of the above
6. Public key cryptography is same as :
 - (A) Private key cryptography
 - (B) Asymmetric cryptography
 - (C) Symmetric cryptography
 - (D) Session key cryptography
7. Hash function are used for :
 - (A) Encryption
 - (B) Decryption
 - (C) Digital signature
 - (D) None of the above
8. PGP stands for:
 - (A) Proper Good Privacy
 - (B) Personal Go Principle
 - (C) Pretty Good Privacy
 - (D) None of these
9. Which of the following monitors all internet and other network activity, looking for suspicious data and preventing unauthorized access ?
 - (A) Intrusion detection system
 - (B) Firewall
 - (C) Data encryption
 - (D) None of the above

10. _____ is used for network level security.
- (A) IPSEC
 - (B) SSL
 - (C) TLS
 - (D) SET
11. Which one of the following is used for creating and verifying the integrity of the message?
- (A) Coding
 - (B) Decryption algorithm
 - (C) Protocol
 - (D) Message Digest
12. For distribution of public key, which of the following uses Trusted third party interface?
- (A) Public announcement
 - (B) Publicly available directory
 - (C) Public key certificate
 - (D) None of these
13. For message authentication, i.e. integrity of message, which mechanism is used?
- (A) MAC
 - (B) IPSEC
 - (C) TLS
 - (D) SET
14. To protect the computer system against the hacker and different kind of viruses, one must always keep _____ on in the computer system.
- (A) Antivirus
 - (B) Firewall
 - (C) Vlc player
 - (D) Script
15. A _____ is an extension of an enterprise's private intranet across a public network such as the internet, creating a secure private connection.
- (A) VPN
 - (B) VAN
 - (C) VDN
 - (D) VSPN
16. Which of the following are famous and common cyber-attacks used by hackers to infiltrate the user's system?
- (A) DDoS and Drive-by Downloads
 - (B) Malware & Malvertising
 - (C) Phishing and Password attacks
 - (D) All of the above
17. Which type of the following malware does not replicate or clone them self's through infection?
- (A) Trojan
 - (B) Bot
 - (C) Worm
 - (D) Viruse

18. Threat on availability of computer resource to its intended users is called:
- (A) Cyber attack
 - (B) Virus attack
 - (C) Worms attack
 - (D) DoS attack
19. PGP encrypts data by using a block cipher called :
- (A) Private data encryption algorithm
 - (B) IDEA
 - (C) RSA
 - (D) Local data encryption algorithm
20. Which is not an objective of network security?
- (A) Secrecy
 - (B) Authentication
 - (C) Access control
 - (D) Lock and unlock
21. The process of verifying the identity of a user :
- (A) Identification
 - (B) Authentication
 - (C) Validation
 - (D) Verification
22. The secure authentication connection in SSL is :
- (A) Chisel
 - (B) Handshaking
 - (C) Tunnel
 - (D) Zerging
23. Examples of hash functions are :
- (A) MD5
 - (B) SDA
 - (C) Both (A) and (B)
 - (D) None of the above
24. Hash function is used in implementing ____ service of network security.
- (A) Availability
 - (B) Integrity
 - (C) Confidentiality
 - (D) All of the above
25. Which software helps the user computer to detect virus :
- (A) Malware
 - (B) Anti virus
 - (C) Adware
 - (D) None of these

26. Which of these refers to verifying the integrity of message ?
- (A) Digital signature
 - (B) Message digest
 - (C) Protocol
 - (D) Decryption algo
27. In system hacking, which of these is most important activity ?
- (A) Information gathering
 - (B) Covering tracks
 - (C) Cracking password
 - (D) None of these
28. Which of these can be considered as the element of cyber security ?
- (A) Application security
 - (B) Operational security
 - (C) Network security
 - (D) All of these
29. In the computer network, the encryption techniques is mainly used for improving :
- (A) Security
 - (B) Performance
 - (C) Reliability
 - (D) Longevity
30. Cryptanalysis is used _____.
- (A) to find some insecurity in a cryptographic scheme
 - (B) to increase the speed
 - (C) to encrypt the data
 - (D) to make new ciphers
31. Which two types of encryption protocols can be used to secure the authentication of computers using IPsec?
- (A) Kerberos V5
 - (B) SHA
 - (C) MD5
 - (D) Both SHA and MD5
32. Hash functions guarantee message integrity and that the message has not been _____.
- (A) Overlooked
 - (B) Replaced
 - (C) Violated
 - (D) Changed
33. Data must arrive exactly as it was sent to receiver from sender, is called _____.
- (A) Message Sending
 - (B) Message Splashing
 - (C) Message Integrity
 - (D) Message Confidentiality

34. IP Security operates in which layer of the OSI model?
- Network
 - Transport
 - Application
 - Physical
35. Which of the following is not a type of symmetric key cryptographic technique ?
- Caesar cipher
 - Data Encryption Standard
 - Playfair cipher
 - Diffie Hellman cipher
36. Which of the following is a passive attack?
- Masquerade
 - Modification of message
 - Distributed Denial of service
 - Traffic analysis
37. A mechanism used to encrypt and decrypt data is known as :
- cryptanalysis
 - cryptology
 - cryptography
 - none of the above
38. Conventional encryption and public key encryption are also _____ called and _____ respectively:
- asymmetric encryption, symmetric encryption
 - symmetric encryption, asymmetric encryption
 - 2 key encryption, one key encryption
 - None of the above
39. Which of the following combination is symmetric encryption techniques ?
- DES, RSA, DIFFIE HELLMAN
 - MD5, SHA_2, DSS
 - IDEA, CAST, TRIPLE DES
 - DSS, IDEA, SHA_1
40. Which is the type of text that is transformed by a cipher at the receiver side ?
- plain text
 - cipher text
 - error text
 - scalar text
41. Which of the following malware does not replicate through infection ?
- rootkit
 - trojan
 - virus
 - worm
42. Which of the following is an independent malicious program that never requires any host program ?
- trojan horse
 - worm
 - trap door
 - virus
43. Why are the factors like confidentiality, authentication, integrity availability considered as fundamental services ?
- To understand hacking process
 - To understand elements of security breach
 - To understand security and its component in better way
 - None of these

44. In order to ensure the security of data , we need to _____ data.
- (A) decrypt
 - (B) encrypt
 - (C) delete
 - (D) zip
45. A communication is said to be insecure where data is transmitted in a manner that allows for interception also called:
- (A) Attack
 - (B) Sniffing
 - (C) ISP
 - (D) Citation
46. The key of key pair used to create digital signature is called:
- (A) Public key
 - (B) Private key
 - (C) Session key
 - (D) Secret key
47. A digital signature uses _____ system of cryptography.
- (A) asymmetric key
 - (B) symmetric key
 - (C) secret key
 - (D) Both (A) and (B)
48. Digital signature scheme can not provide :
- (A) Confidentiality
 - (B) Authentication
 - (C) Integrity
 - (D) None of these
49. _____ is used to create digital signature.
- (A) Public key of receiver
 - (B) Public key of sender
 - (C) Private key of sender
 - (D) Private key of receiver
50. _____ is used to verify digital signature.
- (A) Public key of receiver
 - (B) Public key of sender
 - (C) Private key of sender
 - (D) Private key of receiver
51. Digital signature provides :
- (A) Authentication
 - (B) Non-repudiation
 - (C) Both (A) and (B)
 - (D) None of these

52. In term of web security threat, impersonation of another user is :
- (A) an active attack
 - (B) passive attack
 - (C) Both (A) and (B)
 - (D) None of these
53. Which one is not a protocol of SSL ?
- (A) Record
 - (B) Handshake
 - (C) Alarm
 - (D) Change cipher
54. Full form of SSL :
- (A) Serial Session Layer
 - (B) Secure Session Layer
 - (C) Secure Socket Layer
 - (D) Secure Secure Layer
55. Which protocol is used to give error warning to peer entity ?
- (A) Record
 - (B) Handshake
 - (C) Alert
 - (D) Change cipher
56. Which protocol is used for changing pending state to current state ?
- (A) Alert
 - (B) Handshake
 - (C) Record
 - (D) Change cipher spec
57. IPSEC is designed to provide security at :
- (A) application level
 - (B) transport layer
 - (C) network layer
 - (D) session layer
58. Malware that Encrypts Files and Demands Payment :
- (A) Spyware
 - (B) Ransomware
 - (C) Adware
 - (D) Rootkit
59. Denial of service attack is a threat to :
- (A) confidentiality
 - (B) authentication
 - (C) availability
 - (D) access control

60. Conventional cryptography is also known as _____ or symmetric-key encryption.
- (A) public key
 - (B) protected key
 - (C) secret-key
 - (D) primary key
61. _____ is the unique value for message or content by applying strong Hash function.
- (A) Decryption
 - (B) Encryption
 - (C) Message digest
 - (D) Private key
62. Cryptography can be divided into _____ types.
- (A) 5
 - (B) 4
 - (C) 3
 - (D) 2
63. When plain text is converted to unreadable format, it is termed as _____.
- (A) Confused text
 - (B) Diffused text
 - (C) Cipher-text
 - (D) Ciphon-text
64. _____ is the process or mechanism used for converting ordinary plain text into garbled non-human readable text & vice-versa.
- (A) Malware Analysis
 - (B) Exploit writing
 - (C) Cryptography
 - (D) Reverse engineering
65. Which of the below are current threats to many organisations?
- (A) Cyber attack
 - (B) Loss of information
 - (C) Unauthorised access
 - (D) All of the above
66. PGP is used in:
- (A) email security
 - (B) ftp security
 - (C) browser security
 - (D) server security
67. IDEA is a :
- (A) symmetric cipher
 - (B) asymmetric cipher
 - (C) authentication algo
 - (D) None of these

68. In client server authentication , the client requests from a KDC a _____ for access to specific asset.
- (A) ticket
 - (B) user
 - (C) token
 - (D) card
69. Message _____ means that data must arrive at the receiver as exactly sent.
- (A) integrity
 - (B) confidentiality
 - (C) authentication
 - (D) None of these
70. _____ Service is that where sender must not deny of sending the message.
- (A) Non repudiation
 - (B) Non availi
 - (C) Confidentiality
 - (D) Availability
71. _____ function creates a message digest of a message.
- (A) Hash
 - (B) Coding
 - (C) Decryption
 - (D) None of these
72. A(n) _____ is a govt. authorised organization that binds a public key to an entity and issues a certificate.
- (A) KDC
 - (B) CA
 - (C) Kerberos
 - (D) None of these
73. The _____ criteria says that it is almost impossible to create a message if message digest is known.
- (A) one wayness
 - (B) strong collison resistancee
 - (C) weak collison
 - (D) None of these
74. The criteria of hash that says we cannot find two messages that has same message digest is :
- (A) one wayness
 - (B) strong collison resistance
 - (C) weak collison resistance
 - (D) None of these
75. Which application level protocol uses few manager controlling a set of agents ?
- (A) HTML
 - (B) SNMP
 - (C) TCP
 - (D) All of these

76. The main difference between SNMPv2, and SNMPv3 :
- (A) classification
 - (B) increased security
 - (C) management
 - (D) integration
77. SNMP means :
- (A) Secure Network Management Protocol
 - (B) Set Network Management Protocol
 - (C) Simple Network Management Protocol
 - (D) None of these
78. SNMP is the framework for managing devices in the internet using :
- (A) TCP/IP
 - (B) UDP
 - (C) SNP
 - (D) SMTP
79. _____ ensures the integrity and security of data that pass over network.
- (A) Penetrating tool
 - (B) Firewall
 - (C) Network security protocol
 - (D) Antivirus
80. SSL primarily focuses on :
- (A) integrity and non repudiation
 - (B) integrity and authenticity
 - (C) authentication and privacy
 - (D) confidentiality and integrity
81. The authentication to be affected by use of asymmetric crypto system and hash function is known as :
- (A) Public key
 - (B) Private Key
 - (C) Digital Signature
 - (D) e- governance
82. The SET protocol is used for :
- (A) credit card payment
 - (B) cheque payment
 - (C) debit card payment
 - (D) None of these
83. In SET protocol, a customer encrypts credit card ot using :
- (A) his private key
 - (B) bank private key
 - (C) bank public key
 - (D) merchant public key

84. In SET, customer sends a purchase order :
- (A) encrypted with his public key
 - (B) in plain text
 - (C) using secure digital signature system
 - (D) using bank public key
85. SET uses _____ for secure payment info and purchase order.
- (A) encryption
 - (B) dual signature
 - (C) MAC
 - (D) hashing
86. In network management system, the predefined policy to control to access to network is called :
- (A) Fault management
 - (B) Security management
 - (C) Active management
 - (D) Passive management
87. Communication between end system is encrypted using a key, known as :
- (A) Session key
 - (B) Temporary key
 - (C) Public key
 - (D) Private key
88. In cryptography, cipher means :
- (A) algo for encryption
 - (B) algo for encryption and decryption
 - (C) encrypted message
 - (D) decrypted message
89. In asymmetric key cryptography, for confidentiality, the private key of — is used.
- (A) receiver
 - (B) sender
 - (C) both sender and receiver
 - (D) None of these
90. In asymmetric key cryptography, for authentication, the public key of — is used.
- (A) receiver
 - (B) sender
 - (C) Both sender and receiver
 - (D) none of these
91. Which is not an asymmetric key cryptography ?
- (A) RSA
 - (B) Deffie Hellman
 - (C) DSS
 - (D) IDEA

92. What is data encryption standard (DES) ?
- (A) Block cipher
 - (B) Stream cipher
 - (C) Asymmetric cipher
 - (D) String cipher
93. Caesar cipher is :
- (A) Propositional cipher
 - (B) Substitution cipher
 - (C) Permutation cipher
 - (D) Transposition cipher
94. In cryptography, changing order of letter in message is :
- (A) Transpositional cipher
 - (B) Substitution cipher
 - (C) Both (A) and (B)
 - (D) Diffie Hellman Algo
95. Cryptanalysis is used to :
- (A) to encrypt data
 - (B) find key or plain text or both of the cryptographic scheme
 - (C) to make new cipher
 - (D) None of these
96. Which of these protocols is used to secure HTTP connection ?
- (A) IPSEC
 - (B) TLS
 - (C) ECN
 - (D) FTP
97. Cryptographic hash function takes an arbitrary block of data and returns :
- (A) Variable size byte stream
 - (B) Variable size bit stream
 - (C) Fixed size bit stream
 - (D) None of these
98. An asymmetric key cipher uses :
- (A) 2 keys
 - (B) 3 keys
 - (C) 1 key
 - (D) 4 keys
99. DES uses :
- (A) 56 bit block, 64 bit key
 - (B) 56 bit block, 56 bit key
 - (C) 64 bit block, 64 bit key
 - (D) 64 bit block, 56 bit key
100. DES means :
- (A) Digital Encryption Standard
 - (B) Data Encoding Standard
 - (C) Data Encryption Standard
 - (D) Digital Encoding Set

Rough Work / रफ कार्य

Example :

Question :

Q.1 (A) ● (C) (D)

Q.2 (A) (B) ● (D)

Q.3 (A) ● (C) (D)

5. Each question carries equal marks. Marks will be awarded according to the number of correct answers you have.
6. All answers are to be given on OMR Answer Sheet only. Answers given anywhere other than the place specified in the answer sheet will not be considered valid.
7. Before writing anything on the OMR Answer Sheet, all the instructions given in it should be read carefully.
3. After the completion of the examination, candidates should leave the examination hall only after providing their OMR Answer Sheet to the invigilator. Candidate can carry their Question Booklet.
9. There will be no negative marking.
10. Rough work, if any, should be done on the blank pages provided for the purpose in the booklet.
11. To bring and use of log-book, calculator, pager & cellular phone in examination hall is prohibited.
12. In case of any difference found in English and Hindi version of the question, the English version of the question will be held authentic.

Impt. On opening the question booklet, first check that all the pages of the question booklet are printed properly. If there is any discrepancy in the question Booklet, then after showing it to the invigilator, get another question Booklet of the same series.

उदाहरण :

प्रश्न :

प्रश्न 1 (A) ● (C) (D)

प्रश्न 2 (A) (B) ● (D)

प्रश्न 3 (A) ● (C) (D)

5. प्रत्येक प्रश्न के अंक समान हैं। आपके जितने उत्तर सही होंगे, उन्हीं के अनुसार अंक प्रदान किये जायेंगे।
6. सभी उत्तर केवल ओ०एम०आर० उत्तर-पत्रक (OMR Answer Sheet) पर ही दिये जाने हैं। उत्तर-पत्रक में निर्धारित स्थान के अलावा अन्यत्र कहीं पर दिया गया उत्तर मान्य नहीं होगा।
7. ओ०एम०आर० उत्तर-पत्रक (OMR Answer Sheet) पर कुछ भी लिखने से पूर्व उसमें दिये गये सभी अनुदेशों को सावधानीपूर्वक पढ़ लिया जाये।
8. परीक्षा समाप्ति के उपरान्त परीक्षार्थी कक्ष निरीक्षक को अपनी OMR Answer Sheet उपलब्ध कराने के बाद ही परीक्षा कक्ष से प्रस्थान करें। परीक्षार्थी अपने साथ प्रश्न-पुस्तिका ले जा सकते हैं।
9. निगेटिव मार्किंग नहीं है।
10. कोई भी रफ कार्य, प्रश्न-पुस्तिका में, रफ-कार्य के लिए दिए खाली पेज पर ही किया जाना चाहिए।
11. परीक्षा-कक्ष में लॉग-बुक, कैल्कुलेटर, पेजर तथा सेल्युलर फोन ले जाना तथा उसका उपयोग करना वर्जित है।
12. प्रश्न के हिन्दी एवं अंग्रेजी रूपान्तरण में भिन्नता होने की दशा में प्रश्न का अंग्रेजी रूपान्तरण ही मान्य होगा।

महत्वपूर्ण: प्रश्नपुस्तिका खोलने पर प्रथमतः जाँच कर देख लें कि प्रश्नपुस्तिका के सभी पृष्ठ भलीभाँति छपे हुए हैं। यदि प्रश्नपुस्तिका में कोई कमी हो, तो कक्षनिरीक्षक को दिखाकर उसी सिरीज की दूसरी प्रश्नपुस्तिका प्राप्त कर लें।